

CONGRUENCES AND DENSITY RESULTS FOR PARTITIONS INTO DISTINCT EVEN PARTS

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ABSTRACT. In this paper, we consider the set of partitions $ped(n)$ which counts the number of partitions of n wherein the even parts are distinct (and the odd parts are unrestricted). Using an algorithm developed by Radu, we prove congruences modulo 192 which were conjectured by Nath [Nat24]. Further, we prove a few infinite families of congruences modulo 24 by using a result of Newman. Also, we prove that $ped(9n + 7)$ is lacunary modulo $2^{k+2} \cdot 3$ and $3^{k+1} \cdot 4$ for all positive integers $k \geq 0$. We further prove an infinite family of congruences for $ped(n)$ modulo arbitrary powers of 2 by employing a result of Ono and Taguchi on the nilpotency of Hecke operators.

1. INTRODUCTION

The partition of a positive integer n is a non-increasing sequence of positive integers whose sum is equal to n . For example, $5+3+2$ is a partition of 10.

If $p(n)$ denotes the number of partitions of a positive integer n , then with the convention that $p(0) = 1$, the generating function of $p(n)$ (due to Euler) is given by

$$\sum_{n=0}^{\infty} p(n)q^n = \frac{1}{(q; q)_{\infty}},$$

where

$$(a; q)_{\infty} := \prod_{n=0}^{\infty} (1 - aq^n),$$

where a and q are complex numbers with $|q| < 1$. Throughout this paper, we set

$$f_k := (q^k; q^k)_{\infty}, \quad \text{for any integer } k \geq 1.$$

Ramanujan [Ram19], [Ram20] and [Ram21] discovered three beautiful congruences for the partition function, namely

$$p(5n + 4) \equiv 0 \pmod{5},$$

$$p(7n + 5) \equiv 0 \pmod{7},$$

and

$$p(11n + 7) \equiv 0 \pmod{11}.$$

The subject then got a lot of interest and many mathematicians over the years have found many more interesting results. Many Ramanujan-Type congruences for other classes of partitions such as l -regular partitions, etc (See for example, [CG13]) have also been found by many mathematicians. In this paper, the object of interest is the class of partitions where even parts are distinct and odd parts are unrestricted.

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